

HAT-P-36b

R-0815

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-36_250311 · created 2026-07-28T13:05:07+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460745.8147 +/- 0.0052 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.121 +/- 0.015
Transit depth (Rp/Rs)^2	1.47 +/- 0.37 [%]
Orbital Inclination (inc)	89.99 +/- 2.42
Airmass coefficient 1 (a1)	0.976 +/- 0.013
Airmass coefficient 2 (a2)	0.022 +/- 0.04
Scatter in the residuals of the lightcurve fit is	1.33 %
Best Comparison Star	#3 - [601, 255]
Optimal Aperture	2.85
Optimal Annulus	3.94
Transit Duration (day)	0.0961 +/- 0.0023

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.121 ± 0.015 vs 0.126 (deviation 0.32σ)
Duration fit vs published	0.0961 d vs 0.0925 d (deviation 1.49σ)
Mid-time O–C	-2.7 min
Depth SNR	7.7
Residual scatter	1.33 %

Light curve

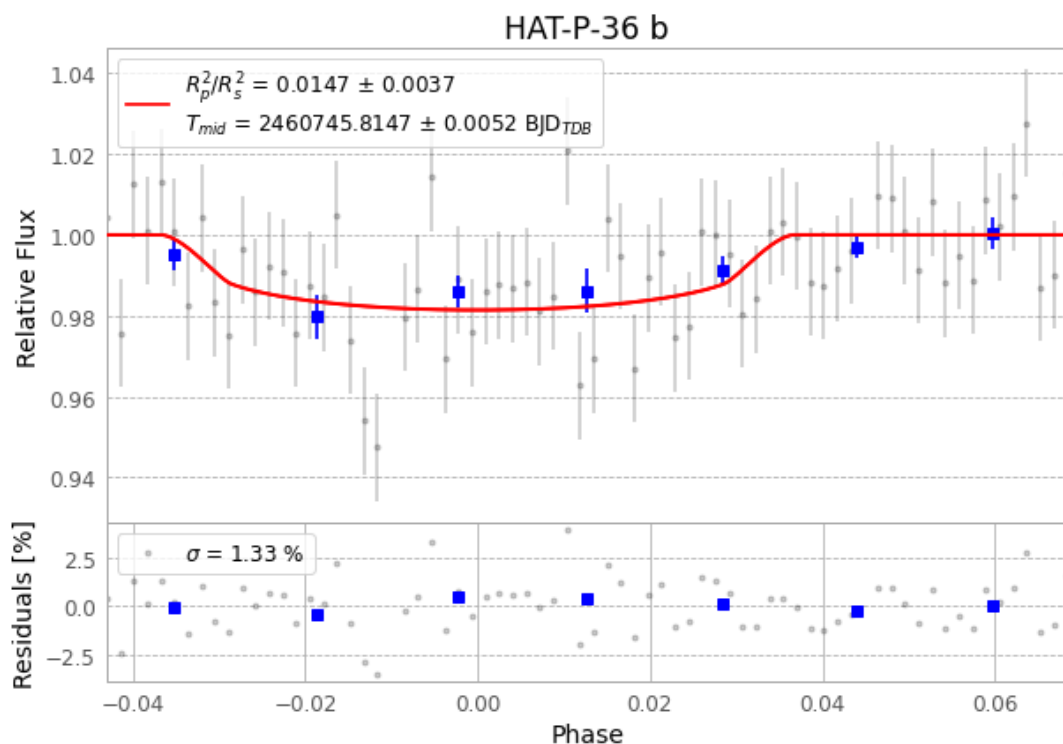


Figure 1 — FinalLightCurve_HAT-P-36 b_2025-03-10.png (SHA-256 9bbb2c9bde8b334a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [376, 230], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 9e7664f1410d75d7...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

72 FITS frames (47 MB), HATP-36250311060316.FITS → HATP-36250311093616.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-250311.log (5ec1a5017814...), FinalLightCurve_HAT-P-36 b_2025-03-10.png (9bbb2c9bde8b...), FinalLightCurve_HAT-P-36 b_2025-03-10.csv (6878305f0820...)

Compute resources

Wall time 110.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 13:05Z.